MAXWELL BLOCK MINIMIZATION AND GAUSSIAN FLUCTUATIONS

Abstract. We construct an Abelian Maxwell block transformation on periodic three-dimensional lattices. The constrained Landau minimizer has exponentially decaying kernels, including first differences and Hölder first differences, uniformly in the mesh and volume. Local axial coordinates make the fluctuation form uniformly positive; its positive inverse square root gives an exact Gaussian block decomposition. We also prove curvature bounds in local real gauges, locality of polynomially weakened kernels, explicit field-domain inclusions, and compatibility of the exact operators with odd periodic covers.

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1. Introduction

Fix an odd integer $L \geq 5$, a power $M = L^m \geq L^2$, and $0 < \alpha < 1$. At level $k \geq 0$, set $\eta = L^{-k}$ and $N = \eta^{-1}$. We use real periodic cochains on $T_{\eta,\ell} = \eta\mathbb{Z}^3/\ell\mathbb{Z}^3$, with pairing $\eta^3 \sum$ on each component. Differences are normalized by the mesh:

$$\partial_\mu f(x) = \eta^{-1}(f(x + \eta e_\mu) - f(x)), \qquad (d\lambda)_\mu = \partial_\mu \lambda, \qquad (d\mathscr{A})_{\mu\nu} = \partial_\mu \mathscr{A}_\nu - \partial_\nu \mathscr{A}_\mu.$$

Write $\delta = d^T$ and $\Delta_{\mathrm{sc}} = \delta d$ on scalars. Distances between sites are periodic sup distances. For bonds use their initial sites, and for coordinate supports use the minimum distance between their points. The local regularity norms $\|B\|_{i,W}$, $i = 1, 2, 3$, mean the supremum, first-difference supremum, and α-Hölder first-difference supremum on W, respectively. Only contained bonds and difference cells enter these local norms. Elsewhere $\|B\|_2$ denotes the quadratic norm; the domain section explicitly uses $\|B\|_i$ for the three global regularity norms. Hölder differences compare parallel output bonds at distance in $(0, 1]$, with the input column and derivative direction fixed.

For $z \in \mathbb{Z}^3/\ell\mathbb{Z}^3$, put $\Delta_z = \{c : |c - z|_\infty < 1/2\}$. The partition M-cubes have side M and centers in $M\mathbb{Z}^3/\ell\mathbb{Z}^3$. The scalar, bond, and plaquette averages to the unit lattice are

$$\begin{aligned}
(\mathbf{Q}_k \lambda)(z) &= \eta^3 \sum_{c \in \Delta_z} \lambda(c), \\
(\mathbf{Q}_k \mathscr{A})_\mu(z) &= \eta^3 \sum_{c \in \Delta_z} \sum_{j=0}^{N-1} \eta \mathscr{A}_\mu(c + j\eta e_\mu), \\
(\mathbf{Q}_k^{(2)} F)_{\mu\nu}(z) &= \eta^3 \sum_{c \in \Delta_z} \sum_{i,j=0}^{N-1} \eta^2 F_{\mu\nu}(c + i\eta e_\mu + j\eta e_\nu).
\end{aligned} \tag{1}$$

All sums are taken on the periodic covering lattice. Let $\mathcal{R}_k$ be the orthogonal projection onto $\Delta_{\mathrm{sc}}(\ker \mathbf{Q}_k)$ in scalar cochains. The Landau representative $H_k^{\mathrm{ph}}A$ minimizes $\|d\mathscr{A}\|_2^2$ under

$$\mathbf{Q}_k\mathscr{A} = A, \qquad \mathcal{R}_k\delta\mathscr{A} = 0. \tag{2}$$

Set

$$\Delta_k = (dH_k^{\mathrm{ph}})^T dH_k^{\mathrm{ph}}. \tag{3}$$

For one block step on the unit lattice, let $B(y) = \{z : |z-y|_\infty < L/2\}$, $y \in L\mathbb{Z}^3/\ell\mathbb{Z}^3$. The scalar average is $\mathbf{Q}\lambda(y) = L^{-3}\sum_{z\in B(y)}\lambda(z)$. On bonds put

$$(\mathbf{Q}A)(y, y+Le_\mu) = L^{-4}\sum_{z\in B(y)} A(\Gamma_{z,z+Le_\mu}), \qquad (\tau A)(y,z) = \tfrac{1}{6}\sum_\pi A(\Gamma^\pi(y,z)).$$

Here $A(\Gamma)$ is the oriented bond sum; the first paths are straight, while the latter change coordinates in each of the six orders. Write $\mathcal{Z} = \{Z : \mathbf{Q}Z = 0,\ \tau Z = 0\}$. For a coarse field A' on side ℓ/L, set $A'_L(b) = L^{-1/2}A'(b/L)$.

Theorem A. Assume that $\ell/(LM)$ is an odd integer at least 13. There are constants $C_P^{\mathrm{ph}} > 0$ and $0 < \gamma_P \le 1$, depending only on L, α, for which the following assertions hold uniformly in k, ℓ.

The minimizer H_k^{ph} exists uniquely, is real linear, and satisfies $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$. It preserves constants and closed fields, and $H_0^{\mathrm{ph}} = 1$. Each of its kernel, first-difference kernel, and short Hölder first-difference kernel is bounded by $C_P^{\mathrm{ph}} e^{-\gamma_P d(b,b')}$. Also

$$|\Delta_k(b,b')| \le C_P^{\mathrm{ph}} e^{-\gamma_P d(b,b')}.$$

There is a real coordinate space V_k with an orthonormal basis $\{\Upsilon\}$, each vector supported in an L-block or on a noncentral face bond, and a bijection $C : V_k \to \mathcal{Z}$. The basis choices are translates of one fixed block basis, and

$$\dim V_k = (2 - 2L^{-3})\ell^3, \qquad \|z\|_2 \le \|Cz\|_2 \le \sqrt{5L^3}\|z\|_2.$$

The form $\widetilde{\Delta}_k = C^T\Delta_k C$ is strictly positive, with upper and lower bounds depending only on L, and

$$|\widetilde{\Delta}_k^{-1/2}(\Upsilon, \Upsilon')| \le C_P^{\mathrm{ph}} e^{-\gamma_P d(\Upsilon,\Upsilon')}.$$

Here the distance on the right is between basis supports.

For every real A', the minimum $A_k^{\min}(A')$ of $\langle A, \Delta_k A\rangle$ under $\mathbf{Q}A = A'_L,\ \tau A = 0$ exists uniquely. With $\mathrm{Sq}_k = C\widetilde{\Delta}_k^{-1/2}$,

$$\begin{aligned}\langle A_k^{\min}(A') + \mathrm{Sq}_k\zeta,\ \Delta_k(A_k^{\min}(A') + \mathrm{Sq}_k\zeta)\rangle \\ = \|dH_{k+1}^{\mathrm{ph}}A'\|_2^2 + \|\zeta\|_2^2.\end{aligned} \tag{4}$$

The operator at level $k+1$ on the right is formed on side ℓ/L and mesh η/L. The harmonic part of A' is arbitrary. The probability law on the affine constraint space with density proportional to $\exp(-\langle A, \Delta_k A\rangle/2)$, relative to its Euclidean volume, is the image of the standard product Gaussian μ_I under $A = A_k^{\min}(A') + \mathrm{Sq}_k\zeta$. For every $q \ge 1$,

$$\mu_I\{\sup_\Upsilon |\zeta(\Upsilon)| \le q\} = e^{-\ell^3\delta\varepsilon^0(q)}, \qquad 0 \le \delta\varepsilon^0(q) \le 12e^{-q^2/2}.$$

If $\square$ is a partition M-cube and $\square^\dagger$ its concentric enlargement of side $(2L+1)M$, then for every real A there is an independent real gauge ν on this nonwrapping enlargement such that

$$\|H_k^{\mathrm{ph}}A - d\nu\|_{i,\square^\dagger} \le C_B\|dA\|_\infty, \qquad i = 1,2,3,$$

where C_B depends only on L, M, α; the subscripts i denote the three regularity norms.

Finally, let $\ell_2 = J\ell_1$, with J odd and both sides as above, and use the same translated block basis. Periodic pullback intertwines $H_k^{\mathrm{ph}}, \Delta_k, C, \widetilde{\Delta}_k, (\widetilde{\Delta}_k + xI)^{-1}$ for $x \geq 0$, $\widetilde{\Delta}_k^{-1/2}, \mathrm{Sq}_k$, and $A_k^{\min}$ on the two tori. ■□

We develop the Abelian block argument of [Cur, Section 10]. The two gauge choices have different uses. The Landau representative admits a Fourier description with constants uniform in the fine mesh. The axial slice has local coordinates on which the curvature energy is uniformly coercive. The inverse square root of this coordinate form gives the Gaussian variables in (4).

The uniformity of the Landau bounds requires cancellation at the zero Fourier alias before the mesh tends to zero. After that cancellation, the remaining alias sums converge on a fixed complex strip. A differentiated summability estimate, using $\alpha < 1$, permits contour shifting for all three kernels. For the fluctuation form, exponential conjugation bounds the resolvent uniformly; its integral gives locality of the positive inverse square root. These arguments are proved in sections 2.2 and 3.2.

The remaining sections provide the photon estimates used in a nonlinear block integration. Curvature controls the reconstructed field after independent local real gauge changes. Polynomial weakening along parameter paths preserves exponential decay and separates linked components; its exact bounds are stated in proposition 5.1. The field-domain theorem, theorem B, states the additional charge inequalities and the margin for complex fluctuations. Only the exact operators are asserted to commute with periodic pullback. No fermion operator or interacting estimate is assumed.

2. The Landau minimizer and locality

2.1. **Coarse averages and the minimizer.** The averages in (1) preserve constants and the mean of a bond field, are onto on bond fields, and satisfy

$$\mathbf{Q}_k d = d\mathbf{Q}_k, \qquad d\mathbf{Q}_k = \mathbf{Q}_k^{(2)} d, \qquad \|\mathbf{Q}_k^{(2)} F\|_2 \leq \|F\|_2. \tag{5}$$

On scalars $\mathbf{Q}_k^T \lambda$ is constant with value $\lambda(z)$ on Δ_z, and $\mathbf{Q}_k \mathbf{Q}_k^T = 1$. The differential identities follow by telescoping along the segments in (1). The plaquette average has total weight one. Jensen's inequality and summation over coarse sites count each fine plaquette with total weight η^3, proving its contraction. To see surjectivity on bonds, give the fine bond leaving a unit block in direction μ the value $NA_\mu(z)$, and give the other bonds value zero. Every segment of length one starting in Δ_z crosses exactly one such boundary bond, and its average is $A_\mu(z)$.

Proposition 2.1. The minimizer exists uniquely. It is real and linear, and $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$. It preserves constant fields; if $dA = 0$, then $dH_k^{\mathrm{ph}} A = 0$. At $k = 0$ it is the identity. Its characterization is

$$\mathbf{Q}_k \mathscr{A} = A, \qquad \delta d \mathscr{A} \in \operatorname{ran} \mathbf{Q}_k^T \text{ on bonds}, \qquad \Delta_{\mathrm{sc}} \delta \mathscr{A} \in \operatorname{ran} \mathbf{Q}_k^T \text{ on scalars}. \tag{6}$$

■

Proof. Surjectivity makes the averaging constraint nonempty. The possible curvatures form an affine subspace of the finite lattice cochain space, on which the squared norm has a minimum. Differentiating along the kernel of $\mathbf{Q}_k$ shows that a minimizer satisfies

$$\langle \delta d \mathscr{A}, B \rangle = 0;$$

this is the middle condition in (6). The parallelogram identity shows that all minimizers have the same curvature.

For a closed periodic bond field the discrete Hodge decomposition is $B = c + d\lambda$, where c is its mean. One can see this directly in Fourier variables: at nonzero momentum, closedness says that the components of $\widehat{B}$ are proportional to the forward-difference symbol, and the proportionality factor is $|\widehat{d}|^{-2} \widehat{d}^* \cdot \widehat{B}$. At zero momentum only the constant remains. Exact fields have mean zero.

Hence a closed $B \in \ker \mathbf{Q}_k$ is $d\lambda$ with $\lambda \in \ker \mathbf{Q}_k$: average the decomposition, use (5), and subtract the constant value of $\mathbf{Q}_k\lambda$. Thus the unconstrained minimizers are $\mathscr{A}_1 + d(\ker \mathbf{Q}_k)$.

The scalar Laplacian is injective on $\ker \mathbf{Q}_k$. Indeed, $\Delta_{\mathrm{sc}}\lambda = 0$ gives $d\lambda = 0$, so λ is constant and its average forces it to be zero. For $\lambda \in \ker \mathbf{Q}_k$,

$$\mathcal{R}_k\delta(\mathscr{A}_1 + d\lambda) = \mathcal{R}_k\delta\mathscr{A}_1 + \Delta_{\mathrm{sc}}\lambda.$$

Since the first term lies in the range $\Delta_{\mathrm{sc}}(\ker \mathbf{Q}_k)$, exactly one such gauge correction imposes the Landau constraint. Moreover $\mathcal{R}_k f = 0$ is equivalent to $\Delta_{\mathrm{sc}} f \perp \ker \mathbf{Q}_k$; this proves the last condition of (6). All these equations are real and linear, so uniqueness proves the reality and linearity of H_k^{ph}.

Constants satisfy the equations. If $A = c + d\lambda$ is closed, then $c + d\mathbf{Q}_k^T\lambda$ is a closed fine field with average A, so the minimum curvature energy is zero. This proves preservation of closed fields. Finally $\mathbf{Q}_0 = 1$ gives the level-zero assertion. □

2.2. **The minimizer symbol and uniform regularity.** The Fourier calculation must separate the zero alias before the continuum mesh limit is taken. Set $h = (N-1)/2$ and, for $s \in \mathbb{C}$,

$$u(s) = N^{-1}\sum_{j=-h}^{h} e^{isj/N}, \qquad s_N(s) = 2N\sin(s/(2N)), \qquad c(s) = 2\sin(s/2).$$

The identity $u(s)s_N(s) = c(s)$ follows by telescoping the geometric sum. For $q \in \mathbb{C}^3$ define

$$d_\mu(q) = N(e^{iq_\mu/N} - 1), \quad d_\mu^*(q) = N(e^{-iq_\mu/N} - 1), \quad D(q) = \sum_\mu s_N(q_\mu)^2, \quad U(q) = \prod_\mu u(q_\mu),$$

and

$$V_\mu(q) = e^{i(1-N^{-1})q_\mu/2}u(q_\mu), \quad V_\mu^*(q) = V_\mu(-q), \quad \rho_\mu(p) = e^{ip_\mu} - 1, \quad \rho_\mu^*(p) = e^{-ip_\mu} - 1.$$

The stars in these formulas denote analytic symbols. At real arguments they are complex conjugates. The aliases are $q_m = p + 2\pi m$, $m \in \{-h, \dots, h\}^3$, and

$$V_\mu(q_m)d_\mu(q_m) = \rho_\mu(p), \qquad V_\mu^*(q_m)d_\mu^*(q_m) = \rho_\mu^*(p), \qquad V_\mu V_\mu^* = u(q_{m,\mu})^2. \tag{7}$$

The scalar averaging symbol is $U(q_m)$; on bonds it is $U(q_m)V_\mu(q_m)$. Thus the Euler equations (6) can be solved separately in each alias class.

Write $D_0 = D(p)$, $U_0 = U(p)$, $u_\mu = u(p_\mu)$, and

$$X_\mu = \sum_{m\neq 0}\frac{U(q_m)^2u(q_{m,\mu})^2}{D(q_m)}, \qquad Y = \sum_{m\neq 0}\frac{U(q_m)^2}{D(q_m)^2},$$
$$N_\mu = U_0^2u_\mu^2 + D_0X_\mu, \qquad N_G = U_0^2 + D_0^2Y,$$
$$\mathcal{N} = \prod_\mu N_\mu, \qquad \Pi = \mathcal{N} - \sum_\mu s_N(p_\mu)^2X_\mu\prod_{\nu\neq\mu}N_\nu.$$

The coarse form symbol and the scalar multiplier for the Landau condition are

$$\begin{aligned}\Lambda_{\mu\nu} &= \delta_{\mu\nu}\frac{D_0}{N_\mu} - \frac{\rho_\mu\rho_\nu^*U_0^2\mathcal{N}}{N_\mu N_\nu\Pi},\\ \omega_\nu &= \frac{\rho_\nu^*D_0U_0^2\mathcal{N}}{N_\nu N_G\Pi},\\ h_{\mu\nu}(p,m) &= \frac{U(q_m)V_\mu^*(q_m)\Lambda_{\mu\nu}}{D(q_m)} + \frac{d_\mu(q_m)U(q_m)\omega_\nu}{D(q_m)^2} \quad (m\neq 0).\end{aligned} \tag{8}$$

The zero-alias formula has its apparent pole cancelled:

$$h_{\mu\nu}(p,0) = \delta_{\mu\nu}\frac{U_0V_\mu^*(p)}{N_\mu} + \frac{d_\mu(p)\rho_\nu^*U_0^3(X_\mu - u_\mu^2D_0Y)\mathcal{N}}{N_\mu N_\nu N_G\Pi}. \tag{9}$$

Indeed, $N_\mu - u_\mu^2 N_G = D_0(X_\mu - u_\mu^2 D_0 Y)$, so (9) equals the preceding formula wherever $D_0 \neq 0$.

For real $p \in [-\pi,\pi]^3$, the nonzero aliases have $D(q_m) > 0$, and $X_\mu, Y \geq 0$. The positivity of u_μ gives $N_\mu, N_G > 0$. If $p \neq 0$, the identity

$$\Pi = \sum_\mu \frac{s_N(p_\mu)^2}{D_0} U_0^2 u_\mu^2 \prod_{\nu\neq\mu} N_\nu \tag{10}$$

shows that $\Pi > 0$; at zero $N_\mu = N_G = \Pi = 1$. The algebraic identity $U_0^2 \sum_\mu \rho_\mu \rho_\mu^*/N_\mu = D_0\Pi/\mathcal{N}$ implies $\sum_\mu \rho_\mu^* \Lambda_{\mu\nu} = 0$. Substitution in the alias equations now gives

$$\widehat{(H_k^{\mathrm{ph}} A)}_\mu(q_m) = \sum_\nu h_{\mu\nu}(p,m)\widehat{A}_\nu(p), \qquad \widehat{(\Delta_k A)}_\mu(p) = \sum_\nu \Lambda_{\mu\nu}(p)\widehat{A}_\nu(p).$$

To check the first formula, average the displayed alias expression. The sums including $m = 0$ are N_μ/D_0 and N_G/D_0^2. Its two terms give $\widehat{A}_\mu$ plus opposite multiples of ρ_μ, so the average is A. Its divergence is $U(q_m)\widehat{\omega}(p)/D(q_m)$; applying the scalar Laplacian gives the scalar range condition, and subtracting its gradient from $\widehat{D(q_m) H_k^{\mathrm{ph}} A}$ gives $U(q_m)V_\mu^*(q_m)\widehat{\Lambda}_\mu(p)$. These are the other two Euler conditions. At $p = 0$ only the zero alias survives, giving the constant field. Uniqueness of the real minimizer proves the formula. Pairing its bond Euler equation with $H_k^{\mathrm{ph}} A'$ and using $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$ identifies the symbol of Δ_k with Λ.

There is a strip of fixed width, independent of N, in which these symbols are analytic. Here is the estimate that provides that width and the derivative bounds. On $|\operatorname{Re} p_\mu| < 5\pi/4$, $|\operatorname{Im} p_\mu| < 1/4$, put $a = (8/(5\pi))\sin(5\pi/8)$ and $c_h = \cosh(1/8)$. The elementary complex sine identities give

$$\operatorname{Re} s_N(s)^2 \geq a^2(\operatorname{Re} s)^2 - c_h^2(\operatorname{Im} s)^2, \qquad |u(s)| \leq c_h$$

in the required alias range. Consequently, for $m \neq 0$,

$$\operatorname{Re} D(q_m) \geq |m|_2^2, \qquad |u(q_{m,\mu})| \leq C\langle m_\mu\rangle^{-1}, \qquad |d_\mu(q_m)| \leq C\langle m_\mu\rangle, \tag{11}$$

where $\langle t\rangle = \max(1,|t|)$. For the first estimate use $|\operatorname{Re} q_{m,\mu}| \geq (3\pi/4)|m_\mu|$ when $m_\mu \neq 0$; the lower bound before simplification is $a^2(3\pi/4)^2|m|_2^2 - 3c_h^2/16 \geq |m|_2^2$. The bound on u follows from $u = c/s_N$, and the bound on d_μ from its sine expression. Thus $X_\mu, Y, N_\mu, N_G, \Pi$ are uniformly bounded on the larger rectangle, since their alias sums are bounded by constant multiples of $\sum_m \prod_\mu \langle m_\mu\rangle^{-2}$.

On its real cube $u_\mu \geq a$, so $N_\mu \geq a^8$, $N_G \geq a^6$, and (10) gives $\Pi \geq a^{24}$. If an analytic function is bounded by B on that rectangle and is at least $c > 0$ on the slightly smaller real cube, apply Schwarz's lemma to $t \mapsto f(p' + itp'') - f(p')$ on the disk $|t| < 4B/c$. For $|p''|_\infty < c/(16B)$ its argument stays in the larger rectangle, and $|f(p) - f(p')| \leq c/2$. Apply this to the three denominators. It gives a fixed width $g_2 > 0$ on $|\operatorname{Re} p_\mu| < 5\pi/4 - 1/4$ where all remain bounded away from zero. In particular, the zero-alias formula is analytic there.

In this smaller strip the alias bounds imply

$$|h_{\mu\nu}(p,m)| \leq C\left(\langle m_\mu\rangle^{-1}|m|_2^{-2} + \langle m_\mu\rangle|m|_2^{-4}\right)\prod_\sigma \langle m_\sigma\rangle^{-1} \quad (m \neq 0), \tag{12}$$

and $|h(p,0)| \leq C$. For every $0 \leq \alpha < 1$ this proves

$$\sup_p \sum_m |h_{\mu\nu}(p,m)| \max\{1, |d_\lambda(q_m)|, |d_\lambda(q_m)||q_m|_1^\alpha\} \leq S_\alpha, \tag{13}$$

with S_α independent of N. In fact, after multiplication by $\langle m\rangle_\infty^{1+\alpha}$, the first term in (12) is bounded by

$$C\langle m_\mu\rangle^{-2}\prod_{\sigma\neq\mu}\langle m_\sigma\rangle^{-(3-\alpha)/2},$$

and the second by $C\prod_\sigma \langle m_\sigma\rangle^{-4/3}$. Both product sums converge; the first is where $\alpha < 1$ is used.

For $x \in \eta\mathbb{Z}^3$, set $K_{\mu\nu}(p;x) = \sum_m e^{iq_m\cdot x} h_{\mu\nu}(p,m)$. It is 2π-periodic in p. To check periodicity, away from $D_0 = 0$ write $E_\mu = N_\mu/D_0$ and $G = N_G/D_0^2$ as sums over all aliases, and $r = \sum_\mu \rho_\mu \rho_\mu^*/E_\mu$. Then

$$\Lambda_{\mu\nu} = \delta_{\mu\nu}/E_\mu - \rho_\mu\rho_\nu^*/(E_\mu E_\nu r), \qquad \omega_\nu = \rho_\nu^*/(E_\nu G r).$$

Translation of p by 2π permutes the alias classes modulo $2\pi N\mathbb{Z}^3$, and every summand, including $e^{iq_m\cdot x}$, is periodic modulo this latter lattice. Thus K and Λ agree on real overlaps of the rectangles; analyticity extends the identity and defines them on the full strip. Define the covering kernel by $H^\infty_{\mu\nu}(x) = (2\pi)^{-3}\int_{[-\pi,\pi]^3} K_{\mu\nu}(p;x)\mathrm{d}p$. Shift the contour to $p + i\gamma_H \operatorname{sgn}(x)$, with $\gamma_H = g_2/2$. The vertical faces cancel by periodicity, and (13) gives

$$|H^\infty(x)|, |\partial H^\infty(x)|, |z|^{-\alpha}|\partial H^\infty(x+z) - \partial H^\infty(x)| \le C S_\alpha e^{-\gamma_H|x|_1}, \qquad x, z \in \eta\mathbb{Z}^3, \quad 0 < |z| \le 1.$$

The last estimate uses $|e^{iq\cdot z} - 1| \le 2.5|z|^\alpha |q|_1^\alpha$ on the shifted contour. The torus kernel is $H_k^{\mathrm{ph}}(b,b') = \sum_{n\in\mathbb{Z}^3} H^\infty(c - z' + \ell n)$, with the corresponding direction indices. Summing the exponential bound over the lifts proves

$$|H_k^{\mathrm{ph}}(b,b')|, |\partial H_k^{\mathrm{ph}}(b,b')|, |\delta_\alpha \partial H_k^{\mathrm{ph}}((b,\tilde b);b')| \le C_H(\alpha) e^{-\gamma_H d(b,b')}. \tag{14}$$

Here $b, \tilde b$ have the same direction and their initial points are at distance at most one. The same contour argument for the bounded symbol Λ gives

$$|\Delta_k(b,b')| \le C_\Delta e^{-\gamma_H d(b,b')}, \qquad \|\Delta_k\|_{2\to 2} \le C_\Delta^{(2)}. \tag{15}$$

All three constants are independent of k and ℓ. For the operator norm, Parseval's formula reduces the estimate to $\sup_p \|\Lambda(p)\|$. A uniform bound A_1 for its nine entries gives $\|\Lambda(p)\| \le 3A_1$, so one may take $C_\Delta^{(2)} = 3A_1$.

3. Axial coordinates and the Gaussian block identity

3.1. **Axial coercivity on a block.** For an interior bond $b = (z, z + e_\nu)$, subtract the two paths from y to its endpoints and append b. For each order this loop bounds the strips belonging to the directions that come after ν. Their total number of plaquettes is $\sum_{\mu \text{ after } \nu} |z_\mu - y_\mu|$. If $\tau A = 0$, averaging over the six orders gives

$$|A(b)| \le \tfrac12 \sum_{\mu\ne\nu} |z_\mu - y_\mu| \sup_{p\subset B(y)} |dA(p)| \le \tfrac{L-1}{2} \sup_{p\subset B(y)} |dA(p)|. \tag{16}$$

For a face between $B(y)$ and $B(y+Le_\nu)$, let $a(z)$ be its face bond values and let p_c be the maximum curvature on plaquettes in these two blocks and across their common face. A transverse face step is one plaquette plus the difference of two interior bond values. Thus $|a(z) - a(z')| \le L|z - z'|_1 p_c$. The defining average $\mathbf{Q}A$ counts each face bond L times and $L^3(L-1)$ interior bond occurrences. Therefore its face mean $\bar a$ satisfies

$$|\bar a| \le L|(\mathbf{Q}A)(c)| + \tfrac12 (L-1)^2 p_c.$$

The average transverse distance from a face site is at most $L-1$, so $|a(z) - \bar a| \le L(L-1)p_c$. Combining these gives

$$|A(b)| \le \tfrac32 L^2 p_c + L|(\mathbf{Q}A)(c)|. \tag{17}$$

For interior bonds use (16) and omit the coarse-average term. In particular, on the axial fluctuation space,

$$\mathbf{Q}Z = 0, \quad \tau Z = 0 \quad\Longrightarrow\quad \|Z\|_\infty \le \tfrac32 L^2 \|dZ\|_\infty, \qquad \|Z\|_2^2 \le 9L^7 \|dZ\|_2^2. \tag{18}$$

For the second inequality, square the local bound and replace each local curvature maximum by its sum of squares. An interior plaquette belongs to the neighborhoods of at most $3L^3 + 3L^2 \le 4L^3$ bonds; a face plaquette to those of at most L^2 bonds. Summation gives $(9/4)L^4 \cdot 4L^3 = 9L^7$.

3.2. **Fluctuation coordinates and the positive square root.** The fluctuations in one block integration satisfy $\mathbf{Q}Z = 0$ and $\tau Z = 0$. We first choose local coordinates on this slice. The positive inverse square root of the resulting quadratic form then gives independent Gaussian variables.

For an L-block $B(y)$, let $\mathrm{Bd}(y)$ denote its interior bond functions, with the counting inner product. The functional τ_y has rank $L^3 - 1$: given values $f(x)$ for $x \neq y$, extend $f(y) = 0$ and apply τ_y to the interior restriction of df. Every path sum telescopes to $f(x)$. Thus

$$\dim \ker \tau_y = 3L^2(L-1) - (L^3 - 1) = 2L^3 - 3L^2 + 1.$$

The three coordinate reflections preserve $\ker \tau_0$ and act as commuting orthogonal involutions. Choose an orthonormal basis e_i^0 of simultaneous eigenvectors and translate it to e_i^y. This choice depends only on L.

Each face between neighboring blocks has a central bond, on the line joining their centers. Let

$$V_k = \bigoplus_y \ker \tau_y \ \oplus \bigoplus_{b \text{ noncentral face bond}} \mathbb{R}\delta_b, \qquad \{\Upsilon\} = \{e_i^y\} \cup \{\delta_b\}.$$

Here δ_b is the point mass on the positively oriented bond b. These are orthonormal coordinates, and

$$\dim V_k = d_V \ell^3, \qquad d_V = 2 - 2L^{-3}. \tag{19}$$

The support of a basis vector lies in one L-block or consists of the two ends of a bond.

For $z \in V_k$ define Cz to agree with z off the central bonds, and set, on the central bond b_c of the face $c = \langle y, y + Le_\mu \rangle$,

$$(Cz)(b_c) = -\sum_{\substack{b \text{ on } c \\ b \neq b_c}} z(b) - \sum_{b \text{ interior}} \frac{n_c(b)}{L} z(b). \tag{20}$$

The second sum is over the bonds of direction μ in the two adjacent blocks; $n_c(b)$ counts the straight averaging paths that contain b. Numbering the bonds on such a line by $t = 0, \dots, 2L-2$ gives $n_c(b) = \min(t+1, 2L-1-t)$. The face bond has multiplicity L; every interior bond has multiplicity at most $L-1$. Consequently

$$L^4(\mathbf{Q}Z)(c) = L \sum_{b \text{ on } c} Z(b) + \sum_b n_c(b) Z(b).$$

Only $Z(b_c)$ is unknown in the equation for this face. Central bonds occur in no axial path and in no other face equation. It follows that C is a real linear bijection

$$C : V_k \longrightarrow \{Z : \mathbf{Q}Z = 0,\ \tau Z = 0\}. \tag{21}$$

The following coordinate estimates will be used both for the covariance and for the small-field domain:

$$\|Cz\|_\infty \le c_C \sup_\Upsilon |z(\Upsilon)|, \qquad \|z\|_2 \le \|Cz\|_2 \le (5L^3)^{1/2} \|z\|_2, \qquad \sum_b |(C\Upsilon)(b)| \le c_1, \tag{22}$$

where $c_C = 3L^{9/2}$ and $c_1 = 14L^{3/2}$. To prove the first bound, normalize the coordinate supremum to one. An interior value is at most $(2L^3)^{1/2}$ by Cauchy–Schwarz, and a noncentral face value is at most one. Formula (20) bounds a central value by $L^2 + 2L^3(2L^3)^{1/2} \le 3L^{9/2}$. For the second bound, the new central values are supported away from z. A central row has at most $2L^3$ entries, of modulus at most one, and an entry contributes to at most two such rows. Cauchy–Schwarz therefore adds at most $4L^3\|z\|_2^2$. For a face coordinate, $C\delta_b$ has two entries of modulus one. For an interior unit vector, the interior ℓ^1 norm is at most $(3L^3)^{1/2}$, and the total contribution of the six adjacent central bonds is at most six times this norm. This proves the last estimate with $c_1 = 14L^{3/2}$. The same formula gives

$$d(x, x') \le L \quad (x \in \operatorname{supp} C\Upsilon,\ x' \in \operatorname{supp} \Upsilon). \tag{23}$$

Coordinate sign changes and translations by $M\mathbb{Z}^3$ permute $\{\pm\Upsilon\}$. For the interior vectors this follows from the simultaneous eigenvector choice; for face coordinates it follows from the preservation

of central bonds. Fix a cutoff $p_{0,k} > 0$; its charge-dependent value is chosen in section 6. The standard Gaussian μ_I on V_k and its normalized restriction $\widehat{\mu}_k$ to

$$\mathrm{Box}_k = \{z : \ |z(\Upsilon)| \le p_{0,k} \text{ for every } \Upsilon\}$$

are invariant under these transformations and under $z \mapsto -z$. Independence of the coordinates gives

$$\mu_I(\mathrm{Box}_k) = \exp(-\ell^3 \delta\varepsilon_k^0), \qquad \delta\varepsilon_k^0 = -d_V \log \Pr(|g| \le p_{0,k}) \le 12 \exp(-p_{0,k}^2/2), \tag{24}$$

when $p_{0,k} \ge 1$, where g is standard normal. Indeed, the Gaussian tail is at most

$$2(2\pi)^{-1/2} p_{0,k}^{-1} \exp(-p_{0,k}^2/2);$$

at $p_{0,k} = 1$ this is less than $1/2$, and $-\log(1-t) \le 2t$ for $t \le 1/2$ suffices.

Set

$$\widetilde{\Delta}_k = C^T \Delta_k C, \qquad C_k = \widetilde{\Delta}_k^{-1}.$$

The first of these definitions makes sense before invertibility has been proved. From the axial estimate, the contraction of the plaquette average, and $\mathbf{Q}_k H_k^{\mathrm{ph}} = 1$, we get

$$\|z\|_2^2 \le \|Cz\|_2^2 \le 9L^7 \|dCz\|_2^2 = 9L^7 \|\mathbf{Q}_k^{(2)} dH_k^{\mathrm{ph}} Cz\|_2^2 \le 9L^7 \langle z, \widetilde{\Delta}_k z\rangle.$$

The upper bound follows from (15) and (22). Thus, with

$$c_P = \max\{9L^7, \ 5L^3 C_\Delta^{(2)}\}, \qquad C_{\mathrm{Dt}} = c_1^2 C_\Delta \exp(2\gamma_H L),$$

we have

$$c_P^{-1} \le \widetilde{\Delta}_k \le c_P, \qquad |\widetilde{\Delta}_k(\Upsilon, \Upsilon')| \le C_{\mathrm{Dt}} \exp(-\gamma_H d(\Upsilon, \Upsilon')). \tag{25}$$

For the second estimate, expand $\langle C\Upsilon, \Delta_k C\Upsilon'\rangle$ as a double bond sum. Every contributing pair satisfies $d(\Upsilon, \Upsilon') \le d(b, b') + 2L$ by (23); the two ℓ^1 bounds then give the asserted constant.

The distance between supports has a triangle defect at most L. We use the following finite-matrix estimate.

Lemma 3.1. Let I be finite, let d be symmetric and nonnegative with $d(a,a) = 0$, and suppose $d(a,c) \le d(a,b) + d(b,c) + \phi$ for some $\phi \ge 0$. Let T be real symmetric, $T \ge t_0 I > 0$, and $|T(a,b)| \le C_T e^{-g d(a,b)}$, where $g > 0$. Choose $S > 0$ such that

$$S \ge e^{g\phi/2} \sup_a \sum_b (d(a,b) + \phi) e^{-g d(a,b)/2}, \qquad \theta = \min\{g/2, t_0/(2C_T S)\}.$$

Then, for $x \ge 0$,

$$|(T + xI)^{-1}(a,c)| \le (t_0/2 + x)^{-1} e^{-\theta d(a,c)}, \qquad |T^{-1/2}(a,c)| \le \sqrt{2/t_0}\, e^{-\theta d(a,c)}.$$

■

Proof. Fix c and conjugate T by the diagonal matrix $W(a,a) = \exp(\theta d(a,c))$. The error $E = WTW^{-1} - T$ satisfies

$$|E(a,b)| \le \theta C_T \exp(g\phi/2)(d(a,b) + \phi) \exp(-g d(a,b)/2),$$

because $|d(a,c) - d(b,c)| \le d(a,b) + \phi$ and $|\exp(u) - 1| \le |u| \exp(|u|)$. The Schur test gives $\|E\| \le t_0/2$. Therefore $\|(T + x + E)^{-1}\| \le (t_0/2 + x)^{-1}$ for $x \ge 0$. Undoing the conjugation yields

$$|(T + x)^{-1}(a,c)| \le (t_0/2 + x)^{-1} \exp(-\theta d(a,c)).$$

The scalar integral $\pi^{-1} \int_0^\infty x^{-1/2}(t + x)^{-1}\, \mathrm{d}x = t^{-1/2}$ and finite-dimensional diagonalization now give

$$T^{-1/2} = \frac{1}{\pi} \int_0^\infty x^{-1/2} (T + x)^{-1}\, \mathrm{d}x, \qquad |T^{-1/2}(a,c)| \le (2/t_0)^{1/2} \exp(-\theta d(a,c)). \tag{26}$$

□

Apply this to $T = \widetilde{\Delta}_k$, $\phi = L$, $t_0 = c_P^{-1}$ and $g = \gamma_H$. The basis support count gives

$$\sup_{\Upsilon} \sum_{\Upsilon'} \exp(-g d(\Upsilon, \Upsilon')) \le c_{\mathrm{bas}}(g), \qquad c_{\mathrm{bas}}(g) = 2L^3 \exp(2gL)(1 + 6/g)^3.$$

For example, assign a basis vector to its block center or to an end of its face bond. At most $2L^3$ vectors are assigned to any site, and replacing support distance by point distance costs at most $2L$; the remaining lattice sum is bounded by $(1 + 6/g)^3$. Since $t \exp(-gt/4) \le 2/g$, we may take

$$S_P = \exp(\gamma_H L/2)(L + 2/\gamma_H) c_{\mathrm{bas}}(\gamma_H/4), \quad \gamma_P = \min\{\gamma_H/2,\ (2c_P C_{\mathrm{Dt}} S_P)^{-1}\}.$$

With

$$C_P^{\mathrm{ph}} = \max\{C_H(\alpha), C_{\mathrm{Dt}}, (2c_P)^{1/2}, 4\exp(2)\},$$

all the kernels in (14), (25), and the positive square root satisfy their respective bounds at rate γ_P with constant C_P^{ph}. In particular,

$$C_k^{1/2} = \widetilde{\Delta}_k^{-1/2}, \qquad |C_k^{1/2}(\Upsilon, \Upsilon')| \le C_P^{\mathrm{ph}} \exp(-\gamma_P d(\Upsilon, \Upsilon')). \tag{27}$$

At $k = 0$, $H_0^{\mathrm{ph}} = 1$; its first differences and their Hölder differences are supported within distance two and have entries of modulus at most two. Our choice $C_P^{\mathrm{ph}} \ge 4\exp(2)$ therefore includes this level as well.

3.3. **The exact block decomposition.** For a real unit bond field A' on the torus of side ℓ/L, put $A'_L(b) = L^{-1/2} A'(b/L)$. Let $A_k^{\min}(A')$ minimize $\langle A, \Delta_k A\rangle$ under

$$\mathbf{Q}A = A'_L, \qquad \tau A = 0.$$

This affine space is nonempty: set each face bond of the face c equal to $LA'_L(c)$, and set every interior bond to zero. Its direction is $C(V_k)$, and (25) makes the quadratic form positive definite on this direction. The minimizer is therefore unique and is characterized by

$$\langle A_k^{\min}(A'), \Delta_k C z\rangle = 0 \quad (z \in V_k). \tag{28}$$

This characterization is a real linear system with a unique solution, so $A_k^{\min}$ is real and linear.

Define $\mathrm{Sq}_k = C C_k^{1/2}$. It maps V_k bijectively onto the axial fluctuations, has zero block average and mean, and

$$\langle \mathrm{Sq}_k \zeta, \Delta_k \mathrm{Sq}_k \zeta'\rangle = \langle \zeta, C_k^{1/2} \widetilde{\Delta}_k C_k^{1/2} \zeta'\rangle = \langle \zeta, \zeta'\rangle.$$

Thus the exact Gaussian decomposition is

$$\langle A_k^{\min}(A') + \mathrm{Sq}_k\zeta, \Delta_k(A_k^{\min}(A') + \mathrm{Sq}_k\zeta)\rangle = \langle A_k^{\min}(A'), \Delta_k A_k^{\min}(A')\rangle + \|\zeta\|_2^2. \tag{29}$$

The density $\exp(-\langle A, \Delta_k A\rangle/2)$ on the affine constraint space becomes a constant times $\exp(-\|\zeta\|_2^2/2)$. The affine change of variables has a constant Jacobian, so normalization gives the standard Gaussian in the ζ coordinates. Restricting it to Box_k retains exactly the mass in (24).

We also need the relation between the two backgrounds. With the scalar scaling $\lambda_L(x) = L^{1/2}\lambda(x/L)$, there is a unique real scalar $\omega_k(A')$ with $\mathbf{Q}_{k+1}\omega_k(A') = 0$ such that

$$H_k^{\mathrm{ph}} A_k^{\min}(A') = (H_{k+1}^{\mathrm{ph}} A')_L - d\omega_k(A'), \qquad \|dH_k^{\mathrm{ph}} A_k^{\min}(A')\|_2 = \|dH_{k+1}^{\mathrm{ph}} A'\|_2. \tag{30}$$

The harmonic part of A' is arbitrary. Here $\mathbf{Q}_{k+1} = \mathbf{Q}\mathbf{Q}_k$ is the average from the mesh-η_k lattice of side ℓ to its L-spaced lattice. The operator H_{k+1}^{ph} is formed at mesh η_{k+1} and side ℓ/L, before scaling by L.

Here is the minimizing argument. At mesh η, the scaled field $(H_{k+1}^{\mathrm{ph}} A')_L$ minimizes the curvature norm under the $(k+1)$-step constraint $\mathbf{Q}_{k+1}\mathscr{A} = A'_L$. This follows by scaling the defining problem;

the curvature L^2 norm is unchanged. The field $\mathscr{B} = H_k^{\mathrm{ph}} A_k^{\min}(A')$ has the same average. If $\mathscr{A}$ is any competitor, set $A = \mathbf{Q}_k \mathscr{A}$. Choose the one-step restricted gauge

$$\mu(x) = (\tau A)(y, x) - L^{-3} \sum_{x' \in B(y)} (\tau A)(y, x'), \qquad x \in B(y).$$

Then $\mathbf{Q}\mu = 0$ and $\tau(A - d\mu) = 0$, by telescoping the exact-field path sums. Therefore

$$\|d\mathscr{A}\|_2^2 \geq \|dH_k^{\mathrm{ph}} A\|_2^2 = \|dH_k^{\mathrm{ph}}(A - d\mu)\|_2^2 \geq \|d\mathscr{B}\|_2^2.$$

Both backgrounds are minimizers of the same problem. The Landau-minimizer argument shows that their difference is the coboundary of a scalar with zero $(k+1)$-step average. This scalar is unique: a closed field with zero average is exact, and the only constant scalar with zero block average is zero. The two backgrounds are real and linear in A', hence so is ω_k. This proves (30) and identifies the first term on the right of (29) with $\|dH_{k+1}^{\mathrm{ph}} A'\|_2^2$.

4. Curvature control in local gauges

Assume from now on that $\ell/(LM)$ is odd and at least 13. For an M-cube $\square$, let $\square^\dagger$ be its enlargement by LM in each direction. Its side is $s_d = (2L+1)M \leq 3\ell/13$. For a bond field B on a set W, write

$$\|B\|_{1,W} = \|B\|_{\infty,W}, \qquad \|B\|_{2,W} = \|\partial B\|_{\infty,W}, \qquad \|B\|_{3,W} = \|\delta_\alpha \partial B\|_{\infty,W}.$$

Only bonds and differences contained in W enter these norms. They are ordinary differences of the photon coefficients; no connection occurs in their definition.

Proposition 4.1. For every real unit bond field A, including an arbitrary constant part, and every M-cube $\square$, there is a real scalar ν on $\square^\dagger$ such that

$$\|H_k^{\mathrm{ph}} A - d\nu\|_{i,\square^\dagger} \leq C_B \|dA\|_\infty, \qquad i = 1, 2, 3. \tag{31}$$

The constant C_B depends on L, M, α and is independent of k, ℓ and the harmonic part of A. ∎

Proof. Let a be the mean of A, put $A^0 = A - a$, and choose the center x_0 of $\square$ as base point. On the canonical lifted period cell

$$W = x_0 + \{-(\ell-1)/2, \ldots, (\ell-1)/2\}^3$$

define $\lambda(x)$ to be the line sum of A^0 along the rectilinear path from x_0 to x, in coordinate order $1, 2, 3$. This defines a scalar on the torus. Put $A'' = A^0 - d\lambda$ and $p_A = \|dA\|_\infty$. The path sums of A'' vanish. For a bond whose two ends lie in W without crossing its boundary, the strip identity used in the axial estimate gives

$$|A''(x, x + e_\mu)| \leq \sum_{\nu > \mu} |x_\nu - x_{0,\nu}| p_A \leq 2d(x, x_0) p_A. \tag{32}$$

Indeed, commuting the final bond through the later portions of the two base-point paths sweeps one strip in each direction $\nu > \mu$. Each strip contributes its plaquette sum, and both remaining path sums are zero.

A seam bond lies on a closed coordinate line with $\ell - 1$ inner bonds. Let $\mathrm{hol}(v)$ be the line sum of A^0 in that direction. Translating this line by one transverse step changes its sum by at most ℓp_A, by summing the intervening ℓ plaquettes. Its mean over the transverse two-torus is zero, since A^0 has mean zero. The mean distance in each coordinate circle is at most $\ell/4$, hence $|\mathrm{hol}(v)| \leq \ell^2 p_A/2$. The line sum of $d\lambda$ around a closed line vanishes. Removing the inner bonds using (32) therefore gives

$$|A''(b')| \leq |\mathrm{hol}(v)| + (\ell - 1)\ell p_A \leq \tfrac{3}{2}\ell^2 p_A \tag{33}$$

for the seam value. In particular this construction also gives a global real gauge with $\|A^0 - d\lambda\|_\infty \leq 2\ell^2 \|dA\|_\infty$.

Since $H_k^{\mathrm{ph}} d\lambda$ is closed and has mean zero, it is exact, say $d\nu'$. On the nonwrapping cube $\square^\dagger$, let x denote lifted coordinates and set $\nu = a \cdot x + \nu'$. Then $H_k^{\mathrm{ph}} A - d\nu = H_k^{\mathrm{ph}} A''$ there. If $b \subset \square^\dagger$ and b' is inner, its initial point is at distance at most $d(b,b')+2$ from an end of b. Thus (32) gives

$$|A''(b')| \le (2d(b,b') + 4 + s_d) p_A.$$

For a seam bond, $d(b,b') \ge (\ell - 1 - s_d)/2$.

Put $g = \gamma_P$ and $c_{\mathrm{bd}}(g) = 3\exp(3g)(1+6/g)^3$, which bounds the sum of $\exp(-g d(b,b'))$ over unit bonds. Summing any of the three kernel bounds against the inner bonds gives

$$C_P^{\mathrm{ph}} p_A \big[(s_d+4) c_{\mathrm{bd}}(g) + (2/g) c_{\mathrm{bd}}(g/2)\big],$$

using $2t\exp(-gt) \le (2/g)\exp(-gt/2)$. For the seam contribution split off half the decay. Since $s_d \le 3\ell/13$,

$$\exp(-g d(b,b')) \le \exp(g/4)\exp(-5g\ell/26)\exp(-g d(b,b')/2), \qquad \ell^2 \exp(-5g\ell/26) \le 15/g^2.$$

Together with (33), this bounds the seam contribution by $62 C_P^{\mathrm{ph}} p_A g^{-2} c_{\mathrm{bd}}(g/2)$. The proposition follows with

$$C_B = C_P^{\mathrm{ph}} \big[(s_d+4) c_{\mathrm{bd}}(\gamma_P) + (2/\gamma_P + 62/\gamma_P^2) c_{\mathrm{bd}}(\gamma_P/2)\big]. \tag{34}$$

□

5. Weakened photon kernels

For $a = M$ or LM, let π_a consist of the closed cubes of half-side a centered at $a\mathbb{Z}^3/\ell\mathbb{Z}^3$. These are overlapping parameter cubes, distinct from the partition M-cubes. Two parameter cubes are linked when they contain a common lattice site; components below refer to chains of such links. For a set S of unit sites, let $\mathrm{Cub}_a(S)$ be all parameter cubes meeting S.

Choose the shortest rectilinear path $\mathrm{Pth}(p,p')$ in coordinate order $1,2,3$. Its signed coordinate offsets are uniquely determined because ℓ is odd. A site belongs to at most 27 parameter cubes. If S is connected by nearest-neighbor steps, then $\mathrm{Cub}_a(S)$ is linked: consecutive sites lie together in a cube, and all cubes containing either site are linked to this cube. Along a coordinate segment, at most nine transverse choices of centers occur. Counting the longitudinal centers for the three segments of $\mathrm{Pth}(p,p')$ gives

$$\#\mathrm{Cub}_a(\mathrm{Pth}(p,p')) \le 9(3d(p,p')/a + 9). \tag{35}$$

An L-block, and the ends of a unit bond, meet at most 27 cubes of either kind.

We smooth only in the row variable of the minimizer. Let

$$\vartheta(r) = \begin{cases} 0 & r \le 0, \\ 3r^2 - 2r^3 & 0 \le r \le 1, \\ 1 & r \ge 1, \end{cases} \qquad \varphi(t) = \vartheta(2(3/4 - |t|)), \qquad \chi_c(x) = \prod_{\nu=1}^{3} \varphi((x_\nu - c_\nu)/M).$$

The coordinates in the last expression are represented with modulus at most $\ell/2$. Put $\chi_c(b) = \chi_c(x_b)$, where x_b is the midpoint of b. The identity $\vartheta(r) + \vartheta(1-r) = 1$ implies $\sum_{j\in\mathbb{Z}} \varphi(t-j) = 1$, and therefore $\chi_c \ge 0$ and $\sum_c \chi_c = 1$ on the torus. Moreover φ' is bounded by three and is Lipschitz with constant 24. Tensor products and the averaged derivative formula for the discrete difference give, for $\tilde b = b + r$ with $|r| \le 1$,

$$|\partial_\nu \chi_c(b)| \le 3/M, \qquad |\chi_c(\tilde b) - \chi_c(b)| \le 9|r|/M, \qquad |\partial_\nu \chi_c(\tilde b) - \partial_\nu \chi_c(b)| \le 42|r|/M^2. \tag{36}$$

For the last bound the derivative factor contributes $24/M^2$, and the two remaining factors contribute $2 \cdot 9/M^2$. If $\chi_c(b) \ne 0$, both ends of b belong to the cube centered at c.

For a unit bond b' with ends p_1', p_2', define

$$P'(c,b') = \mathrm{Cub}_M(\mathrm{Pth}(c,p_1') \cup \mathrm{Pth}(c,p_2')), \qquad w(b,b';s) = \sum_c \chi_c(b) \prod_{a \in P'(c,b')} s_a.$$

For an interior basis vector set $\mathrm{St}(e_i^y) = B(y)$ and $\mathrm{Pt}(e_i^y) = \{y\}$; for δ_b, both sets are the two ends of b. Put

$$P''(\Upsilon,\Upsilon') = \mathrm{Cub}_{LM}\left(\mathrm{St}(\Upsilon)\cup\mathrm{St}(\Upsilon')\cup\bigcup_{\substack{p\in\mathrm{Pt}(\Upsilon)\\ p'\in\mathrm{Pt}(\Upsilon')}}\mathrm{Pth}(p,p')\right).$$

The weakened kernels are

$$\begin{aligned} H_k^{\mathrm{ph}}(s)(b,b') &= H_k^{\mathrm{ph}}(b,b')w(b,b';s), \quad (k\ge 1),\\ C_k^{1/2}(s)(\Upsilon,\Upsilon') &= C_k^{1/2}(\Upsilon,\Upsilon')\prod_{a\in P''(\Upsilon,\Upsilon')} s_a, \quad (k\ge 0). \end{aligned} \tag{37}$$

The two families have parameter sets π_M and π_{LM}, respectively. When both occur in one expression, s denotes their pair. Each entry is a polynomial, independent of the connection and real for real parameters. At $s=1$ the exact operators are recovered. For general complex parameters, the second family is defined by the displayed kernel; positivity or the square-root identity is used only at $s=1$.

Fix $0<\alpha_0\le 1/4$ and take

$$M = L^m \ge L^2, \qquad 2\alpha_0 n_P \log M \le \gamma_P M, \qquad n_P = 402. \tag{38}$$

Parameters range over the closed polydisc $|s_a|\le M^{\alpha_0}$. Such an M exists because $(\log M)/M\to 0$. If $\chi_c(\hat b)\neq 0$ for a translate of b by distance at most two, then $d(c,x_b)\le M+1$; applying (35) to both paths gives

$$\#P'(c,b') \le 234 + 54 d(b,b')/M \le n_P(1+d(b,b')/M).$$

At most 27 such centers occur: in each coordinate their representatives lie in an interval of length $3M/2+5<3M$. The set P'' has two endpoint sets and at most four paths. Since their endpoints are within L of the respective supports, the same count gives

$$\#P''(\Upsilon,\Upsilon') \le 378 + 216/M + 108 d(\Upsilon,\Upsilon')/(LM) \le n_P(1+d(\Upsilon,\Upsilon')/M).$$

For either product, with corresponding distance D, we have

$$\left|\prod_{a\in P} s_a\right| \le M^{\alpha_0 n_P}\exp(\alpha_0 n_P D\log M/M) \le M^{\alpha_0 n_P}\exp(\gamma_P D/2). \tag{39}$$

This is the loss that the exact exponential decay absorbs.

Proposition 5.1. Under (38), put $\gamma^P = \gamma_P/2$ and $C^P = 116 C_P^{\mathrm{ph}} M^{\alpha_0 n_P}$. The three kernels of $H_k^{\mathrm{ph}}(s)$ satisfy

$$|H_k^{\mathrm{ph}}(s)(b,b')|, |\partial H_k^{\mathrm{ph}}(s)(b,b')|, |\delta_\alpha \partial H_k^{\mathrm{ph}}(s)((b,\tilde b);b')| \le C^P\exp(-\gamma^P d(b,b')), \tag{40}$$

and

$$|C_k^{1/2}(s)(\Upsilon,\Upsilon')| \le C^P\exp(-\gamma^P d(\Upsilon,\Upsilon')). \tag{41}$$

These bounds hold on the closed polydiscs. ∎

Proof. The root estimate follows at once from (27) and (39). For the minimizer, fix b' and abbreviate $H(b) = H_k^{\mathrm{ph}}(b,b')$, $w(b) = w(b,b';s)$ and $b^+ = b+\eta e_\nu$. Put

$$\mathrm{Pm} = M^{\alpha_0 n_P}\exp(\gamma_P d(b,b')/2).$$

For $b, b^+, \tilde b, \tilde b^+$, and the intervening translates, (36) and the 27-center count give

$$|w|\le \mathrm{Pm}, \quad |\partial_\nu w|\le 9\mathrm{Pm}, \quad |w(\tilde b^+) - w(b^+)|\le 27|r|\mathrm{Pm}, \quad |\partial_\nu w(\tilde b) - \partial_\nu w(b)|\le 14|r|\mathrm{Pm}.$$

Here $\eta \le 1/3$ and $M \ge 9$; all translates are within two of the initial point. A rectilinear path between b and $\tilde b = b + r$ has at most $3|r|/\eta$ steps, at distance at most one from b. Summing the first-derivative bound for H along this path gives

$$|H(\tilde b) - H(b)| \le 3|r| C_P^{\mathrm{ph}} \exp(\gamma_P) \exp(-\gamma_P d(b, b')).$$

Use the discrete product rule

$$\partial_\nu (Hw)(b) = (\partial_\nu H)(b) w(b^+) + H(b) \partial_\nu w(b).$$

It bounds the zeroth and first kernels with factors one and ten. Subtracting this formula at b and $\tilde b$ gives four terms: the difference of $\partial_\nu H$, the difference of $w(b^+)$, the difference of H, and the difference of $\partial_\nu w$. Their bounds have factors $1, 27, 27\exp(\gamma_P), 14$, respectively, times $|r|^\alpha C_P^{\mathrm{ph}} \mathrm{Pm} \exp(-\gamma_P d(b, b'))$. We used $|r| \le |r|^\alpha$ and $\gamma_P \le 1$. Their sum is at most 116 times this expression, proving (40). □

At level zero the row smoothing is replaced by the diagonal family

$$H_0^{\mathrm{ph}}(s)(b, b') = \delta_{bb'} \prod_{a \in \pi_M : b \subset a} s_a. \tag{42}$$

Every product has at most 27 factors and at least one. The first difference has two diagonal terms; the Hölder difference has four, supported within distance two. Thus their entries are bounded by $4\exp(2\gamma^P) M^{27\alpha_0} \exp(-\gamma^P d(b, b'))$, which is at most the right side of (40) by $C_P^{\mathrm{ph}} \ge 4\exp(2)$.

The weakening also separates components exactly. Suppose that $s_a = 0$ outside a set $\mathscr{Y}$ of parameter cubes, and let Y_1 be one linked component of their union. A linked path set P meeting this component either lies entirely in it, or contains a cube outside $\mathscr{Y}$ and its parameter product vanishes. Both P' and P'' are linked and contain every parameter cube meeting the respective column support; P'' also contains every cube meeting the row support. In the first family, every contributing row cube c contains the fine bond b and belongs to $P'(c, b')$. It follows that a row or column supported in Y_1 depends only on parameters in that component and has no nonzero entry at an index outside it. A nonzero row is supported in some component. The diagonal family has the same properties because all cubes containing b are linked through an end of b.

Finally, coordinate sign changes preserve the ordered shortest paths, and translations by the relevant center lattice preserve the parameter cubes. The exact kernels commute with these transformations: the averaging, axial and Landau constraints are preserved, and uniqueness identifies the transformed minimizers and covariance. The basis transforms by a signed permutation. The parameter products and χ_c transform accordingly. Hence $H_k^{\mathrm{ph}}(s)$ is covariant under coordinate sign changes and M translations, and $C_k^{1/2}(s)$ under coordinate sign changes and LM translations, with the parameters transformed as well.

6. Domains for block integration

We make explicit the domain conditions used with these operators. Take $0 < \varepsilon \le 1/200$, $e_k \in (0, 1)$, and put $t_{i,k} = e_k^{-3/4 + i\varepsilon}$ for $i = 1, 2, 3$. The set R_k consists of complex fine bond fields with $\|B\|_i < t_{i,k}$ in the three norms. A field belongs to $c\widetilde R_k$, $0 < c \le 1$, if it has a decomposition $B = B_0 + B_1$, with $B_1 \in cR_k$ and B_0 real, such that on each $\square^\dagger$ the field B_0 is gauge equivalent by a real scalar to a real field with norms strictly below $ct_{i,k}$. The decomposition and gauges need only exist. For $\widetilde R_k^-$ the local real representative has the bounds $t_{i,k}$, while the global term B_1 has the bounds $(1 - e_k^\varepsilon) t_{i,k}$. Finally,

$$\mathrm{Dom}_k = \{A \text{ real unit bond field} : \ H_k^{\mathrm{ph}} A \in \tfrac{1}{4} \widetilde R_k\}.$$

These definitions permit every real harmonic part.

Let $C_w \geq C^P$ and $0 < \gamma_w \leq \gamma^P$, and choose c_γ to bound both sums

$$\sup_{\Upsilon} \sum_{\Upsilon'} \exp(-\gamma_w d(\Upsilon, \Upsilon')), \qquad \sup_{b} \sum_{b'} \exp(-\gamma_w d(b, b'))$$

over basis elements and unit input bonds, respectively; in the second supremum b is a fine output bond. The counts above make c_γ independent of k, ℓ. Fix $p, p_0 > 0$ and let $p_{0,k} = (-\log e_k)^{p_0}$ and $p_k = (-\log e_k)^p$. Assume

$$\begin{aligned} p_{0,k} &\leq e_k^{-\varepsilon/4}, \qquad 2C_w^2 c_\gamma^2 c_C e_k^{\varepsilon/4} \leq 1, \\ 8C_B p_k &\leq t_{3,k}, \qquad 8e_k^{3/4-7\varepsilon/2} \leq 1, \qquad L^{3/4} \geq 8/3. \end{aligned} \tag{43}$$

Since the constants are fixed before the charge, these are conditions on an upper bound e_0, valid for every $0 < e_k \leq e_0$.

Theorem B. With the geometry and weakened operators of the preceding sections, assume (38) and (43). Then every real unit bond field with $\|dA\|_\infty \leq p_k$ belongs to Dom_k. For every $\zeta \in \mathrm{Box}_k$ and every pair of parameters in the closed polydiscs, the field

$$\mathscr{Z}_k(s, \zeta) = H_k^{\mathrm{ph}}(s) C C_k^{1/2}(s) \zeta$$

has all three norms bounded by $\frac{1}{2} e_k^{-\varepsilon/2}$. If ℓ/L is also an admissible side and $e_{k+1} = L^{1/2} e_k \in (0, 1)$, then

$$A_k^{\min}(A') + \mathrm{Sq}_k \zeta \in \mathrm{Dom}_k \quad (A' \in \mathrm{Dom}_{k+1},\ \zeta \in \mathrm{Box}_k). \tag{44}$$

If $A \in \widetilde{R}_k^-$ and $t \in \mathbb{C}$, then $A + t\mathscr{Z}_k(s, \zeta) \in \widetilde{R}_k$ whenever

$$|t| \leq \mathrm{rad}_1 := \tfrac{1}{2} e_k^{-3/4+9\varepsilon/2}. \tag{45}$$

■□

Proof. The first assertion follows from (31): its local real representative has each norm at most $C_B p_k \leq t_{3,k}/8 < t_{i,k}/4$. Use $B_1 = 0$ in the domain decomposition.

For the fluctuation bound, the root estimate and the first summation bound give

$$\sup_{\Upsilon} |(C_k^{1/2}(s)\zeta)(\Upsilon)| \leq C_w c_\gamma e_k^{-\varepsilon/4}.$$

The coordinate estimate for C holds also for complex coordinates: evaluating its real bound on the signs of the real row coefficients gives $\sum_\Upsilon |C(b, \Upsilon)| \leq c_C$. Each norm of $H_k^{\mathrm{ph}}(s)v$ is bounded by $C_w c_\gamma \|v\|_\infty$, by the respective kernel sum. Consequently all three norms of $\mathscr{Z}_k(s, \zeta)$ are at most

$$C_w^2 c_\gamma^2 c_C e_k^{-\varepsilon/4} \leq \tfrac{1}{2} e_k^{-\varepsilon/2}.$$

This proves the complex weakened assertion at every level, including the diagonal family at level zero.

For (44), write $\mathscr{A}' = H_{k+1}^{\mathrm{ph}} A'$. It is real and belongs to $\frac{1}{4}\widetilde{R}_{k+1}$. On any coarse enlarged cube there is a real gauge in which $\mathscr{A}'$ is represented by a real field A^c with i-th norm below $t_{i,k+1}/2$: add the real global summand to the local representative. For any fine M-cube $\square$, choose a coarse cube $\square'$ with $\square \subset L\square'$. Then $\square^\dagger \subset L(\square'^\dagger)$. The scaling of the three norms is $L^{-1/2}, L^{-3/2}, L^{-3/2-\alpha}$, respectively, and

$$t_{i,k+1} = L^{-3/8+i\varepsilon/2} t_{i,k}.$$

Thus $(A^c)_L$ has i-th norm strictly below $\frac{1}{2} L^{-7/8+3\varepsilon/2} t_{i,k} \leq \frac{1}{2} L^{-3/4} t_{i,k} \leq 3t_{i,k}/16$. By (30), the new background on $\square^\dagger$ is gauge equivalent to $(A^c)_L + \mathscr{Z}_k(1, \zeta)$. The latter fluctuation is real, and its norms satisfy

$$\tfrac{1}{2} e_k^{-\varepsilon/2} \leq t_{3,k}/16 \leq t_{i,k}/16$$

by (43). The sum has norms strictly below $t_{i,k}/4$. This is the required local representative, again with global summand zero.

Finally, for the radius in (45), the three norms of $t\mathscr{Z}_k(s,\zeta)$ are at most $\frac{1}{4}e_k^{-3/4+4\varepsilon} < e_k^{\varepsilon}t_{3,k}$. This fits the margin in the global summand of $\widetilde{R}_k^-$, and proves the asserted inclusion. The fluctuation is polynomial in the two parameter families and linear in ζ. □

7. Consistency on odd covers

Let $\ell_2 = J\ell_1$ with J odd and with both sides admissible. Write pj* for periodic pullback, also on the coarse unit lattices and on fluctuation coordinates. The common reflection eigenbasis in each block makes the last pullback well defined. Its image is the subspace of ℓ_1-periodic fields, and

$$\langle \mathrm{pj}^* f, \mathrm{pj}^* h\rangle_{\ell_2} = J^3\langle f, h\rangle_{\ell_1}.$$

All exact photon operators commute with this pullback:

$$O^{(2)}\mathrm{pj}^* = \mathrm{pj}^* O^{(1)}, \qquad O = H_k^{\mathrm{ph}}, \Delta_k, C, \widetilde{\Delta}_k, (\widetilde{\Delta}_k + x)^{-1}, C_k^{1/2}, \mathrm{Sq}_k. \tag{46}$$

The superscript specifies the torus; $x \geq 0$. Also

$$(A_k^{\min})^{(2)}(\mathrm{pj}^* A') = \mathrm{pj}^*(A_k^{\min})^{(1)}(A'). \tag{47}$$

To prove these identities, first note that d, δ, $\Delta_{\mathrm{sc}}, \mathbf{Q}_j, \mathbf{Q}_j^T, \tau, C$ have the same finite row formulas on the periodic lifts. They commute with pullback and with translations by $\ell_1\mathbb{Z}^3$. Pulling back the three Euler equations (6) therefore characterizes the larger-torus minimizer with the pulled-back datum. Uniqueness gives the identity for H_k^{ph}.

The exact operators intertwine the orthogonal translations by $\ell_1\mathbb{Z}^3$. Transposing this relation shows that their adjoints also intertwine these translations and preserve the periodic subspaces. If $O^{(2)}\mathrm{pj}^* = \mathrm{pj}^* O^{(1)}$, pairing with two pullbacks and canceling J^3 gives the same identity for O^T. Products give the identities for Δ_k and $\widetilde{\Delta}_k$. Invertibility gives the resolvent identity; integrating it by (26) gives the root identity, and hence the identity for Sq_k.

Finally let $A = (A_k^{\min})^{(1)}(A')$ and test its pullback against a larger-torus axial fluctuation $\widetilde{Z}$. Sum the translates of $\widetilde{Z}$ over $\ell_1\mathbb{Z}^3/\ell_2\mathbb{Z}^3$ to obtain pj$^*Z$. Its average and axial sums vanish, so $Z \in C(V_k)$ on the smaller torus. The field $\Delta_k^{(2)}\mathrm{pj}^* A$ is periodic; consequently

$$\langle \mathrm{pj}^* A, \Delta_k^{(2)}\widetilde{Z}\rangle_{\ell_2} = J^{-3}\langle \mathrm{pj}^*\Delta_k^{(1)} A, \mathrm{pj}^* Z\rangle_{\ell_2} = \langle A, \Delta_k^{(1)} Z\rangle_{\ell_1} = 0.$$

Its average and axial constraints also pull back. The characterization (28) proves (47).

Proof of theorem A. Proposition 2.1 gives the minimizer and its algebraic properties. The alias estimates give (14) and (15). The coordinate construction gives (21), (19) and (22); its positive form and root satisfy (25) and (27). Equations (29) and (30) give (4), with Gaussian restriction mass (24). Proposition 4.1 gives the local real gauges. The exact cover identities are (46) and (47). □

References

[Cur] "The field-strength Osterwalder–Schrader axioms for QED in three dimensions". Comprehensive edition, 263 pp., 20 October 2026, before the modular integration revision. 2026 (cit. on p. 3).